

Nepal Engineering College

Digital Signal Analysis and Processing

Tutorial I

1. Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x_a(t) = 3\cos(5at + \pi/6)$

(b) $x[n] = 3\cos[0.25n]$

(c) $x[n] = \cos[n/8]\cos[\pi n/8]$

(d) $x[n] = \cos[\pi n/2] - \sin[\pi n/8] + 3\cos[\pi n/4 + \pi/3]$

(e) $x[n] = 1 + e^{(j4\pi n)/7} - e^{(j2\pi n)/5}$

1. if $x[n] = \{1, 3, 1, 2, \underset{\uparrow}{2}, -4, 3, 1\}$. Plot $x[n]$, $x[2n-1]$, $x[-2n+3]$.
2. An analog electrocardiogram (ECG) signal contains useful frequencies up to 100 Hz.
 - a. What is the Nyquist rate for this signal?
 - b. Suppose that we sampled this signal at a rate of 150 samples per second, what is the highest frequency that can be represented uniquely at this sampling rate?
3. An analog signal $x_a(t) = 2\sin(480\pi t) - 5\sin(920\pi t)$ is sampled 800 times per second.
 - a. Determine the Nyquist sampling rate for $x_a(t)$.
 - b. Determine the folding frequency.
 - c. What are the frequencies, in radians, in the resulting discrete time signal $x[n]$?
 - d. What is the analog signal $y_a(t)$ you can reconstruct from the samples if the ideal interpolation is used?
4. A digital communication link carries binary-coded words representing samples of an input signal $x_a(t) = 2\sin^2 300\pi t + 5\sin 500\pi t$. The link is operated at 10,000 bits/sec and each input sample is quantized into 1024 different voltage levels.
 - a. What is the sampling frequency and the folding frequency?
 - b. What is the Nyquist rate for the signal $x_a(t)$?
 - c. What are the different frequencies in the resulting discrete-time signals $x[n]$?
 - d. What is the resolution Δ ?
5. Determine the bit rate and the resolution in the sampling of a seismic signal with dynamic range of 0.1 volt if the sampling rate is $F_s = 20$ samples/s and we use an 8-bit ADC. What is the maximum frequency that can be present in the resulting digital seismic signal?
6. Check for the memory, linearity, time invariance, causality and stability of following systems:
 - (a) $y[n] = 4nx[n]$
 - (b) $y[n] = 2x[n^2]$
 - (c) $y[n] = x^2[n-a]$
 - (d) $y[n] = n[x[n]]^2$
 - (e) The accumulator System
7. Consider an LTI system with impulse response $h[n]$ and input $x[n]$. Determine the output of the system, $y[n]$ for the following pairs of inputs and impulse responses.
 - a. $x[n] = \{-2, \mathbf{1}, 2, b\}$; $h[n] = \{a, -2, \mathbf{1}, 3\}$ (Bold no. is the position of $n=0$)
 - b. $x[n] = 2u[n] - u[n-3] - u[n-6]$; $h[n] = 0.8^n u[n]$(Try also the deconvolution from above problems)
8. Given $h_1[n] = \{1, 4, \mathbf{2}, 1, -2\}$ and $h_2[n] = \{1, -\mathbf{2}, 3\}$. Find the overall response $h[n]$ if the two systems are cascade interconnected. Also find the step response of this system. (Bold no. is the position of $n=0$)